

Program: Diploma in Mechatronics	
Course Code : 5432	Course Title: Automation and Control
Semester: 5	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To identify the significance of automation.
- To provide a basic knowledge about the structure and application of power semiconductor devices.
- To develop PLC ladder programs for real time applications.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Electronic Components, Rectifiers	2031	Fundamentals of Electrical and Electronics Engineering	2
Semiconductor Physics	2041	Basic Electronics	2
Amplifiers, Logic Circuits	3031	Analog and Digital Circuits	3

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Understand the concept of Industrial Automation and its applications	13	Understanding
CO2	Illustrate the structure and operation of Power Semiconductor Devices.	16	Understanding
CO3	Explain the working of Controlled Rectifiers and Inverters.	16	Understanding
CO4	Develop PLC ladder program for an application.	13	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand the concept of Industrial Automation and its applications		
M1.01	Describe the need for automation	3	Understanding
M1.02	Explain the types of automation	3	Understanding
M1.03	Explain the elements of an automated system	3	Understanding
M1.04	Illustrate the levels of automation	4	Understanding
Contents: Automation - need for automation - advantages and disadvantages of automation – ten strategies for automation - types of automation - fixed, programmable, flexible and integrated automation – advantages and disadvantages - basic elements of an automated system – levels of automation - future challenges of automation.			
CO2	Illustrate the structure and operation of Power Semiconductor Devices		
M2.01	Illustrate the structure and characteristics of SCR	4	Understanding
M2.02	Explain the turn-on methods for SCR	4	Understanding
M2.03	Illustrate the commutation techniques of SCR	4	Understanding
M2.04	Illustrate the structure and characteristics of DIAC and TRIAC	4	Understanding
	Series Test 1	1	

Contents:

SCR - structure - characteristics - working principle – applications - two transistor analogy – turn on/triggering methods - gate triggering methods - ‘R’ triggering - RC triggering - UJT triggering - commutation techniques - forced commutation circuits (class A to E) DIAC, TRIAC- structure - working principle – characteristics – applications.

CO3	Explain the working of Controlled Rectifiers and Inverters		
M3.01	Illustrate the working of single-phase controlled rectifiers	6	Understanding
M3.02	Describe the operation of inverter circuits	6	Understanding
M3.03	Explain the working of single-phase dual converter	4	Understanding

Contents:

Single phase converters - half wave - full wave midpoint and bridge - working principle with R, RL loads - Basic inverter circuit - working principle - series and parallel inverter circuits - working principle – single phase dual converters - working principle - applications.

CO4	Develop PLC ladder program for an application		
M4.01	Explain the architecture of PLC	3	Understanding
M4.02	List the advantages and applications of PLC	2	Understanding
M4.03	Demonstrate ladder diagram instruction sets of PLC.	4	Understanding
M4.04	Develop ladder program for various applications	4	Applying
	Series Test	1	

Contents:

PLC - architecture - advantages - applications - ladder logic - ladder diagram instruction sets - bit instructions - timer/counter instructions - compare instructions - move instructions - math instructions - program control instructions - ladder programs – logic gates - staircase lamp – conveyor control – water level control – servo motor speed control.

Text / Reference:

T/R	Book Title/ Author
T1	Industrial automation and mechatronics - A.K.Gupta, S.K.Arora
T2	Industrial Electronics and Control - S K Bhattacharya, S Chatterjee
T3	Programmable logic controllers - Frank D Petruzella.
R1	Industrial Electronics and Control - Biswanath Paul - PHI
R2	Power Electronics - P S Bimbhra
R3	Introduction to Programmable Logic Controllers - Gary Dunning - 3rd Edition - Delmar

Online Resources:

SI No	Website Link
1	https://nptel.ac.in/courses/108105062
2	https://nptel.ac.in/courses/108105063
3	https://www.electrical4u.com/electrical-engineering-articles/power-electronics/
4	https://nptel.ac.in/courses/108/102/108102145/